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Subject: Programming with Java	
ASSIGNMENT No : 3	
TITLE : Method Overloading and Static Members	

Problem Statement

Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.

Learning Objectives

To implement method overloading and use static variables and methods in Java programs.

Programme

```
public class Library {

    static int totalBooksIssued = 0;


    void calculateFine(int daysLate) {

        totalBooksIssued++;

        double fine = daysLate * 5;

        System.out.println("Fine (days late only) = Rs." + fine);

    }


    void calculateFine(int daysLate, boolean isReferenceBook) {

        totalBooksIssued++;

        double fine = daysLate * 5;

        if (isReferenceBook) {

            fine += 50;
```

```
        }
        System.out.println("Fine (with reference book charge) = Rs." + fine);
    }

    void calculateFine(int daysLate, boolean isReferenceBook, double damageCharge) {
        totalBooksIssued++;
        double fine = daysLate * 5;
        if (isReferenceBook) {
            fine += 50;
        }
        fine += damageCharge;
        System.out.println("Fine (with damage charge) = Rs." + fine);
    }

    static void displayTotalBooks() {
        System.out.println("\nTotal Books Issued = " + totalBooksIssued);
    }

    public static void main(String args[]) {
        Library l = new Library();

        l.calculateFine(4);
        l.calculateFine(6, true);
        l.calculateFine(3, false, 100);

        Library.displayTotalBooks();
```

```
    }
}
```

Output

```
Fine (days late only) = Rs.20.0
Fine (with reference book charge) = Rs.80.0
Fine (with damage charge) = Rs.115.0

Total Books Issued = 3

=== Code Execution Successful ===
```

Conclusion

This program demonstrates method overloading and the use of static variables and static methods in Java. The overloaded `calculateFine()` methods show how the same method name can handle different parameter combinations, while the static variable `totalBooksIssued` tracks data shared across all method calls. The static method `displayTotalBooks()` confirms that static members can be accessed directly through the class name, without creating an object.